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WORK MACHINE AND CONTROL SYSTEM FOR WORK MACHINE

Abstract

A work machine includes a lower traveling body; an upper turning body mounted on the lower traveling body so as to freely turn; a sound collecting device provided on the upper turning body so as to be able to detect a direction of a sound around the work machine; and a control device configured to report a direction of speech based on the direction of the sound detected by the sound collecting device, when a person existing around the work machine is speaking.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-037494, filed on Mar. 11, 2024, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

Technical Field

[0002] The present invention relates to a work machine and a control system for the work machine.

Description of Related Art

[0003] In the related art, workers often work around an excavator. Therefore, in the case of an excavator, a technique for detecting a person existing around the excavator based on an image captured by an imaging device mounted on an upper turning body is known.

SUMMARY

[0004] According to an embodiment of the present invention, there is provided a work machine including a lower traveling body; an upper turning body mounted on the lower traveling body so as to freely turn; a sound collecting device provided on the upper turning body so as to be able to detect a direction of a sound around the work machine; and a control device configured to report a direction of speech based on the direction of the sound detected by the sound collecting device, when a person existing around the work machine is speaking.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a side view of a work machine according to an embodiment;

[0006] FIG. **2** is a top view of a work machine according to the embodiment;

[0007] FIG. **3** is a diagram illustrating an example of the configuration of an external sound collecting device and an information transmission device attached to the work machine illustrated in FIG. **1**;

[0008] FIG. **4** is a diagram schematically illustrating an example of the configuration of the work machine according to the embodiment;

[0009] FIG. **5** is a top view of the inside of the driving room according to the embodiment;

[0010] FIG. **6** is a conceptual diagram illustrating a two-way conversation between an operator seated in the work machine according to the embodiment and a worker who exists around the work machine;

[0011] FIG. **7** is a diagram illustrating an example of a display screen displayed by a display device according to the embodiment;

[0012] FIGS. **8**A and **8**B are diagrams explaining an example of correcting the direction of speech in an output control part according to the embodiment;

[0013] FIG. **9** is a diagram illustrating an example of a display screen displayed by the display device according to the embodiment;

[0014] FIG. **10** is a diagram illustrating an example of a display screen displayed by the display device according to the embodiment;

[0015] FIG. **11** is a flowchart illustrating a processing procedure for displaying a direction of speech by a controller according to the embodiment;

[0016] FIG. **12** is a top view of another configuration example of a work machine according to another embodiment; and

[0017] FIG. **13** is a schematic view illustrating a configuration example of a remote support system for a work machine according to further another embodiment.

DETAILED DESCRIPTION

[0018] When working in a work machine, there is a case where a person who exists near the work machine speaks to the person in the work machine. The sound detected by a microphone installed in the work machine is output from the speaker to the operator of the work machine. Thus, the operator can recognize the spoken content. However, because the operator of the work machine does not know the direction in which the person who spoke exists, there is a case where the operator of the work machine does not know how to respond.

[0019] Therefore, it is desirable to provide a technology for facilitating the identification of the person who spoke by reporting the direction of the speech and facilitating communication with the person who spoke.

[0020] According to an aspect of the present invention, communication with a person who has spoken is facilitated.

[0021] Embodiments of the present invention will be described below with reference to the drawings. The embodiments described below are exemplary rather than limiting the present invention, and all features and combinations thereof described in the embodiments are not necessarily essential to the present invention. In each of the drawings, the same or corresponding configurations are denoted by the same or corresponding reference numerals, and explanations may be omitted.

[0022] The work machine **100** according to the embodiment of the present disclosure is an excavator. The work machine **100** may be a machine other than an excavator such as a crane, an asphalt finisher, or a forklift. In the illustrated example, the excavator serving as the working machine **100** is an excavator provided with a bucket **6** as an end attachment, but it may be an application machine such as a forestry machine provided with an end attachment other than the bucket **6**.

[0023] In the following description, the work machine **100** is an excavator provided with a bucket **6** as an end attachment, but the present invention is not limited to an excavator. The work machine **100** may be an application machine such as a forestry machine having an end attachment other than the bucket **6**.

First Embodiment

[0024] First, an outline of the work machine **100** will be described with reference to FIGS. **1** and **2**. FIG. **1** is a side view of the work machine **100**, and FIG. **2** is a top view of the work machine **100**.

[0025] The work machine **100** is provided with a lower traveling body **1**, an upper turning body **3** mounted on the lower traveling body **1** so as to be able to turn freely through a turning mechanism **2**, an attachment AT for performing various kinds of work, and a driving room **10**. The front side of the work machine **100** (the upper turning body **3**) corresponds to the direction in which the attachment AT extends with respect to the upper turning body **3** when the work machine **100** is viewed from the top (top view) along the turning axis of the upper turning body **3**. The left side and the right side of the work machine **100** (the upper turning body **3**) correspond to the left side and the right side respectively, as viewed from the operator sitting on the driving seat in the driving room **10**.

[0026] The lower traveling body **1** includes, for example, a pair of left and right crawlers **1**C. Specifically, the crawlers **1**C include a left crawler **1**CL and a right crawler **1**CR. The left crawler **1**CL is driven by a left traveling hydraulic motor **2**ML, and the right crawler **1**CR is driven by a right traveling hydraulic motor **2**MR. The left traveling hydraulic motor **2**ML is a traveling driving part that drives the left crawler **1**CL that is the driven part and can rotate the left crawler **1**CL. The right traveling hydraulic motor **2**MR is a traveling driving part that drives the right crawler **1**CR that is the driven part and can rotate the right crawler **1**CR. The traveling driving part may be an electric motor.

[0027] The upper turning body **3** turns with respect to the lower traveling body **1** when the turning mechanism **2** is driven by the turning hydraulic motor **2**A. The turning hydraulic motor **2**A is a turning driving part that drives the upper turning body **3** that is a driven part and can change the orientation of the upper turning body **3**. The turning driving part may be an electric motor.

[0028] A boom **4** is rotatably attached to the front center of the upper turning body **3**, an arm **5** is rotatably attached to the tip of the boom **4**, and a bucket **6** is rotatably attached to the tip of the arm **5**. In the illustrated example, the boom **4**, the arm **5**, and the bucket **6** constitute an excavation attachment which is an example of the attachment AT. The boom **4**, the arm **5**, and the bucket **6** are driven by a boom cylinder **7**, an arm cylinder **8**, and a bucket cylinder **9**, respectively.

[0029] The bucket **6** is an example of a work tool (end attachment). The bucket **6** is used, for example, for excavation work. Other work tools may be attached to the tip of the arm **5** in place of the bucket **6** depending on the work content. Other work tools may be other types of buckets such as large buckets, slope buckets, and dredging buckets. The other work tools may be types of work tools other than a bucket, such as agitators, breakers, grapples, or lifting magnets.

[0030] The turning hydraulic motor **2**A, the left traveling hydraulic motor **2**ML, the right traveling hydraulic motor **2**MR, the boom cylinder **7**, the arm cylinder **8**, and the bucket cylinder **9** are hydraulic actuators driven by hydraulic oil discharged from the hydraulic pump.

[0031] In the work machine **100**, all or part of the driven parts such as the lower traveling body **1**, the upper turning body **3**, the boom **4**, the arm **5**, and the bucket **6** may be electrically driven. That is, the work machine **100** may be a hybrid excavator or an electric excavator in which all or part of the driven parts are driven by an electric actuator.

[0032] An information transmission device G**1**, an external sound collecting device M**1**, an imaging device S**6**, and an external sound output device SP**1** are attached to the work machine **100**.

[0033] The imaging device S**6** is provided in the upper turning body **3** or the driving room **10**, and captures the surrounding area of the work machine **100** and acquires image information representing the surrounding area of the work machine **100**. In the illustrated example, the imaging device S**6** includes a front camera S**6**F, a left camera S**6**L, a right camera S**6**R, and a rear camera S**6**B.

[0034] The front camera S**6**F is a camera for capturing an area in front of the work machine **100**, and is mounted on the outside of the driving room **10**, such as the roof of the driving room **10** or the side face of the boom **4**. The front camera S**6**F may be mounted, for example, on the ceiling of the driving room **10**, that is, inside of the driving room **10**. The left camera S**6**L is a camera for capturing an area to the left of the work machine **100**, the right camera S**6**R is a camera for capturing an area to the right of the work machine **100**, and the rear camera S**6**B is a camera for capturing an area to the rear of the work machine **100**. Specifically, the front camera S**6**F, the left camera S**6**L, the right camera S**6**R, and the rear camera S**6**B are all monocular wide-angle cameras equipped with imaging devices such as CCD or CMOS, and output the captured images to the display device D**1**. Information of the images captured by the imaging device S**6** is taken into the controller **30**.

[0035] In the illustrated example, the front camera S**6**F is mounted on the roof of the driving room **10**, the left camera S**6**L is mounted on the left end of the upper surface of the upper turning body **3**, the right camera S**6**R is mounted on the right end of the upper surface of the upper turning body **3**, and the rear camera S**6**B is mounted on the rear end of the upper surface of the upper turning body **3**.

[0036] The imaging device S**6** may constitute an object detection device for detecting objects around the work machine **100**. The object detection device may be constituted by a device other than a camera. For example, the object detection device may be a LiDAR. The LiDAR is a device that can measure the distance between a point group of 1 million or more points within the monitoring range and a LiDAR (laser source). The object detection device may be another device that can measure the distance to the object, such as a stereo camera, a distance image camera, or a millimeter-wave radar. When a millimeter-wave radar or the like is used as the object detection device, the object detection device may transmit a large number of signals (such as laser beams) toward the object and receive the reflected signals to derive the distance and direction of the object. Alternatively, the object detection device may be a combination of two or more types of devices. For example, the object detection device may be a combination of an imaging device and a LiDAR, a combination of an imaging device and a millimeter-wave radar, or a combination of an imaging device and a stereo camera.

[0037] The external sound collecting device M**1** is a device that collects external sound and is also referred to as a microphone. In the illustrated example, the external sound collecting device M**1** is provided in the upper turning body **3** or the driving room **10**, and converts sound (air vibration) generated around the work machine **100** into mechanical vibration, and converts the mechanical vibration into electrical signals. Specifically, the external sound collecting device M**1** includes a front microphone M**1**F, a left microphone M**1**L, a right microphone M**1**R, and a rear microphone M**1**B.

[0038] The front microphone M**1**F is a microphone that collects sound generated in front of the work machine **100**, and is mounted on the outside of the driving room **10**, such as the roof of the driving room **10** and the side surface of the boom **4**. The front microphone M**1**F may be mounted, for example, on the ceiling of the driving room **10**, that is, inside of the driving room **10**. The left microphone M**1**L collects sound generated in the left side of the work machine **100**, the right microphone M**1**R collects sound generated on the right side of the work machine **100**, and the rear microphone M**1**B collects sound generated at the rear of the work machine **100**. The electric signals generated by the front microphone M**1**F, the left microphone M**1**L, the right microphone M**1**R, and the rear microphone M**1**B are taken into the controller **30**.

[0039] In the illustrated example, the front microphone M**1**F is mounted on the roof of the driving room **10**, the left microphone M**1**L is mounted on the left end of the upper surface of the upper turning body **3**, the right microphone M**1**R is mounted on the right end of the upper surface of the upper turning body **3**, and the rear microphone M**1**B is mounted on the rear end of the upper surface of the upper turning body **3**. Thus, the four external sound collecting devices M**1** (the front microphone M**1**F, the left microphone M**1**L, the right microphone M**1**R, and the rear microphone M**1**B) are provided at different positions of the upper turning body **3**. Therefore, the controller **30** can detect the direction that is toward the sound source based on the difference (for example, differences in volume) among the sounds collected by the four external sound collecting devices M**1**. When an array microphone is used as the external sound collecting device M**1**, the direction in which the sound source is located can be detected based on, for example, a phase shift or a difference in volume.

[0040] In the illustrated example, each of the four external sound collecting devices M**1** and each of the four imaging devices S**6** are arranged so as to correspond to each other. Specifically, the front microphone M**1**F is arranged so as to be adjacent to the front camera S**6**F, the left microphone M**1**L is arranged so as to be adjacent to the left camera S**6**L, the right microphone M**1**R is arranged so as to be adjacent to the right camera S**6**R, and the rear microphone M**1**B is arranged so as to be adjacent to the rear camera S**6**B.

[0041] The external sound output device SP**1** outputs sound toward the surrounding area of the work machine **100**. In the illustrated example, the external sound output device SP**1** is a non-directional speaker and is configured so as to output sound uniformly in all directions. However, the external sound output device SP**1** may be a directional speaker which outputs sound toward the front.

[0042] The information transmission device G**1** informs the outside of the work machine **100** of the state of the work machine **100**. In the illustrated example, the information transmission device G**1** is provided in the upper turning body **3** or the driving room **10**, and is configured so as to be able to transmit the state of the work machine **100** to the worker around the work machine **100**. Specifically, the information transmission device G**1** includes a front light bar G**1**F, a left light bar

G**1**L, a right light bar G**1**R, and a rear light bar G**1**B.

[0043] The front light bar G**1**F is a light emitting device capable of visually transmitting information to the worker or the like in front of the work machine **100**, and is attached to the outside of the driving room **10**, such as the roof of the driving room **10** or the side of the boom **4**. The front light bar G**1**F may be attached, for example, to the ceiling of the driving room **10**, that is, to the inside of the driving room **10**. The left light bar G**1**L is a light emitting device capable of visually transmitting information to the worker or the like at the left side of the work machine **100**, the right light bar G**1**R is a light emitting device capable of visually transmitting information to the worker or the like at the right side of the work machine **100**, and the rear light bar G**1**B is a light emitting device capable of visually transmitting information to the worker or the like at the rear of the work machine **100**. Each of the front light bar G**1**F, the left light bar G**1**L, the right light bar G**1**R, and the rear light bar G**1**B emits light in response to an electric signal from the controller **30**. In the illustrated example, the light emitting device is an LED light, but other light emitting devices such as a halogen lamp may be used. The light emitting device is a multicolor light emitting type, but may be a monochromatic light emitting type.

[0044] In the illustrated example, the front light bar G**1**F is attached to the roof of the driving room **10**, the left light bar G**1**L is attached to the left end of the upper surface of the upper turning body **3**, the right light bar G**1**R is attached to the right end of the upper surface of the upper turning body **3**, and the rear light bar G**1**B is attached to the rear end of the upper surface of the upper turning body **3**. Thus, the four information transmission devices G**1** (the front light bar G**1**F, the left light bar G**1**L, the right light bar G**1**R, and the rear light bar G**1**B) are provided at different positions of the upper turning body **3**. Therefore, by operating each of the four information transmission devices G**1** separately, the controller **30** can convey the state of the work machine **100** to the workers at the front, left, right, and rear areas of the work machine **100**.

[0045] In the illustrated example, each of the four information transmission devices G**1** and each of the four external sound collecting devices M**1** are arranged so as to correspond to each other. Specifically, the front light bar G**1**F is arranged so as to be adjacent to the front microphone M**1**F, the left light bar G**1**L is arranged so as to be adjacent to the left microphone M**1**L, the right light bar G**1**R is arranged so as to be adjacent to the right microphone M**1**R, and the rear light bar G**1**B is arranged so as to be adjacent to the rear microphone M**1**B.

[0046] FIG. **3** is a diagram illustrating an example of the configuration of the external sound collecting device M**1** and the information transmission device G**1** attached to the work machine **100**. Specifically, FIG. **3** is a perspective view of the left microphone M**1**L and the left light bar G**1**L attached to the substantially rectangular parallelepiped housing. Although the following description referring to FIG. **3** relates to the combination of the left microphone M**1**L and the left light bar G**1**L, the same applies to the combination of the front microphone M**1**F and the front light bar G**1**F, the combination of the right microphone M**1**R and the right light bar G**1**R, and the combination of the rear microphone M**1**B and the rear light bar G**1**B.

[0047] As illustrated in FIG. **3**, the left microphone M**1**L and the left light bar G**1**L are arranged on the left side of the substantially rectangular parallelepiped housing so as to face the area to the left side of the work machine **100**. With this arrangement, the left microphone M**1**L can efficiently collect sounds generated on the left side of the work machine **100**, and the left light bar G**1**L can efficiently convey the state of the work machine **100** to the worker on the left side of the work machine **100**. For example, the left microphone M**1**L can capture the voice spoken by the worker on the left side of the work machine **100**, and the left light bar G**1**L can convey to the worker that the left microphone M**1**L has captured the voice of the worker by emitting light in a predetermined color. In this case, the worker on the left side of the work machine **100** who has spoken to the left microphone M**1**L can confirm that his or her voice has reached the left microphone M**1**L (i.e., the operator of the work machine **100**) by seeing the left light bar G**1**L emitting light in a predetermined color.

[0048] The information transmission device G**1** may be provided at the top of each of the four sides of the driving room **10**. For example, the information transmission device G**1** may be configured such that the front light bar G**1**F is attached to the top of the front face of the driving room **10**, the left light bar G**1**L is attached to the top of the left face of the driving room **10**, the right light bar G**1**R is attached to the top of the right face of the driving room **10**, and the rear light bar G**1**B is attached to the top of the rear face of the driving room **10**. Further, the information transmission device G**1** may be a rotating lamp such as a NICO TORCH attached to the top of the driving room **10**, or a display device such as a liquid crystal display or an organic EL display.
[0049] The controller **30** is an example of a control device and is composed of a computer including, for example, a CPU, a volatile storage device, a nonvolatile storage device, and various input/output interfaces. Then, the controller **30** implements various functions by, for example, reading a program from the nonvolatile storage device, loading the program into the volatile storage device, and causing the CPU to execute the program. In the illustrated example, the controller **30** is configured to implement various functions and control the work machine **100**. The various functions include, for example, a machine guidance function for guiding the manual operation of the work machine **100** by an operator. The various functions may include a contact avoidance function for automatically or autonomously operating or stopping the work machine **100** in order to avoid contact between the work machine **100** and an object existing within the monitoring range around the work machine **100**.
[0050] The boom angle sensor S**1** detects the boom angle which is the rotation angle of the boom **4** with respect to the upper turning body **3**. The arm angle sensor S**2** detects the arm angle which is the rotation angle of the arm **5** with respect to the boom **4**. The bucket angle sensor S**3** detects the bucket angle which is the rotation angle of the bucket **6** with respect to the arm **5**.
[0051] Each of the boom angle sensor S**1**, the arm angle sensor S**2**, and the bucket angle sensor S**3** may be, for example, a rotary encoder, an acceleration sensor, a 6-axis sensor, an IMU (inertial measurement unit), etc., a potentiometer using a variable resistor, a cylinder stroke sensor for detecting the stroke amount of a hydraulic cylinder, etc.
[0052] A detection signal corresponding to the boom angle acquired by the boom angle sensor S**1**, a detection signal corresponding to the arm angle acquired by the arm angle sensor S**2**, and a detection signal corresponding to the bucket angle acquired by the bucket angle sensor S**3** are taken into the controller **30**.
[0053] The body inclination sensor S**4** detects the inclined state of the body (the lower traveling body **1** or the upper turning body **3**) with respect to the horizontal plane. The body inclination sensor S**4** is attached to the upper turning body **3**, for example, and detects the inclination angle of the work machine **100** (that is, the upper turning body **3**) around two axes respectively in the longitudinal direction and the lateral direction. The body inclination sensor S**4** may be, for example, an acceleration sensor, a 6-axis sensor, or an IMU. The detection signal corresponding to the inclination angle detected by the body inclination sensor S**4** is taken into the controller **30**.
[0054] The turning sensor S**5** outputs information related to the turning of the upper turning body **3**. The turning sensor S**5** detects, for example, the turning angle speed of the upper turning body **3** relative to the lower traveling body **1**. The turning sensor S**5** may detect the turning angle. The turning sensor S**5** may be, for example, a gyro sensor, a resolver, or a rotary encoder. A detection signal corresponding to the turning angle or turning angle speed of the upper turning body **3** detected by the turning sensor S**5** is taken into the controller **30**.
[0055] The positioning device PS measures the position of the upper turning body **3**. The positioning device PS may be, for example, a GNSS (Global Navigation Satellite System) compass, and detects the position and orientation of the upper turning body **3**. A detection signal corresponding to the position and orientation of the upper turning body **3** is taken into the controller **30**. The function of detecting the orientation of the upper turning body **3** may be implemented by a direction sensor attached to the upper turning body **3**.

[0056] The driving room **10** is a cabin where the operator is seated, and is mounted on the front left side of the upper turning body **3**. However, the driving room **10** may be omitted in a case where the work machine **100** is operated remotely or in a case where the work machine **100** is operated by fully automatic operation.
[0057] The communication device T**1** communicates with external equipment through a communication network including a mobile communication network, a satellite communication network, or an Internet network. The communication device T**1** may be, for example, a mobile communication module corresponding to a mobile communication standard such as LTE (Long Term Evolution), 4G (4th Generation), 5G (5th Generation), etc., or a satellite communication module for connecting to a satellite communication network.
[0058] The work machine **100** operates actuators in response to the operation of an operator seated in the driving room **10** to drive driven parts such as the lower traveling body **1**, the upper turning body **3**, the boom **4**, the arm **5**, and the bucket **6**.
[0059] Alternatively, the work machine **100** may be configured to be remotely operated from the outside of the work machine **100**. When the work machine **100** is remotely operated, the inside of the driving room **10** may be in an unmanned state.
[0060] Further, the work machine **100** may automatically operate the actuator regardless of the operation of the operator. Thus, the work machine **100** implements the function of automatically operating at least a part of the driven parts such as the lower traveling body **1**, the upper turning body **3**, the boom **4**, the arm **5**, and the bucket **6**, that is, what is referred to as a “machine control function”.
[0061] FIG. **4** schematically illustrates an example of the configuration of the work machine **100**. In FIG. **4**, the mechanical power transmission system, the hydraulic oil line, the pilot line, and the electric control system are indicated by double lines, bold solid lines, bold dashed lines, and dotted lines, respectively.
[0062] The driving system of the work machine **100** includes an engine **11**, a regulator **13**, a main pump **14**, and a control valve unit **17**. The hydraulic driving system of the work machine **100** includes hydraulic actuators such as the turning hydraulic motor **2**A, the left traveling hydraulic motor **2**ML, the right traveling hydraulic motor **2**MR, the boom cylinder **7**, the arm cylinder **8**, and the bucket cylinder **9**.
[0063] The engine **11** is an example of a power source of the work machine **100** and is mounted, for example, at the rear of the upper turning body **3**. The power source of the work machine **100** may be a combination of a power source such as a battery or a fuel cell and an electric motor. Specifically, the engine **11** constantly rotates at a predetermined target rotation speed under direct or indirect control by the controller **30** to drive the main pump **14** and the pilot pump **15**. The engine **11** is, for example, a diesel engine using light oil as fuel. The engine **11** may be a gasoline engine or a hydrogen engine.
[0064] The regulator **13** controls the discharge amount of the main pump **14**. For example, the regulator **13** controls the discharge amount of the main pump **14** by adjusting the angle (tilt angle) of the swash plate of the main pump **14** in response to a control instruction from the controller **30**.
[0065] The main pump **14**, for example, like the engine **11**, is mounted at the rear of the upper turning body **3** and supplies hydraulic oil to the control valve unit **17** through a hydraulic oil line. In the illustrated example, the main pump **14** is a variable displacement hydraulic pump.
[0066] The control valve unit **17** is a hydraulic control device for controlling the hydraulic system in the work machine **100**. In the illustrated example, the control valve unit **17** includes control valves **171** to **176**. The control valve unit **17** is configured to selectively supply hydraulic oil discharged from the main pump **14** to one or more hydraulic actuators through the control valves **171** to **176**. The control valves **171** to **176** control the flow rate of hydraulic oil flowing from the main pump **14** to the hydraulic actuator and the flow rate of hydraulic oil flowing from the hydraulic actuator to the hydraulic oil tank. The hydraulic actuator includes the boom cylinder **7**,

the arm cylinder **8**, the bucket cylinder **9**, the left traveling hydraulic motor **2**ML, the right traveling hydraulic motor **2**MR, and the turning hydraulic motor **2**A. Specifically, the control valve **171** corresponds to the left traveling hydraulic motor **2**ML, the control valve **172** corresponds to the right traveling hydraulic motor **2**MR, and the control valve **173** corresponds to the turning hydraulic motor **2**A. The control valve **174** corresponds to the bucket cylinder **9**, the control valve **175** corresponds to the boom cylinder **7**, and the control valve **176** corresponds to the arm cylinder **8**.

[0067] The pilot pump **15** is an example of a pilot pressure generating device, and is configured to supply hydraulic oil to the hydraulic control equipment via a pilot line. In the illustrated example, the pilot pump **15** is a fixed-capacity hydraulic pump. However, the pilot pressure generating device may be implemented by the main pump **14**. That is, the main pump **14** may have a function of supplying hydraulic oil to the control valve unit **17** via a hydraulic oil line and a function of supplying hydraulic oil to various hydraulic control equipment via a pilot line. In this case, the pilot pump **15** may be omitted.

[0068] The discharge pressure sensor **28** is configured to detect the discharge pressure of the main pump **14**. In the example illustrated in the figure, the discharge pressure sensor **28** outputs the detected value to the controller **30**.

[0069] An operation device **26** is a device used by an operator to operate an actuator. The operation device **26** includes, for example, an operating lever and an operating pedal. The actuator may be a hydraulic actuator or an electric actuator.

[0070] The operating sensor **29** is configured to detect the operation contents of an operator using the operation device **26**. In the present embodiment, the operating sensor **29** detects the operating direction and the operating amount of the operation device **26** corresponding to each of the actuators, and outputs the detected values to the controller **30**. In the illustrated example, the controller **30** can control the opening area of the proportional valve **31** according to the output of the operating sensor **29**. Then, the controller **30** supplies the hydraulic oil discharged from the pilot pump **15** to the pilot port of the corresponding control valve in the control valve unit **17**. The pressure (pilot pressure) of the hydraulic oil supplied to each of the pilot ports is, in principle, the pressure corresponding to the operating direction and the operating amount of the operation device **26** corresponding to each of the hydraulic actuators. Thus, the operation device **26** is configured to supply the hydraulic oil discharged from the pilot pump **15** to the pilot port of the corresponding control valve in the control valve unit **17**.

[0071] The proportional valve **31**, which functions as a control valve for machine control, is arranged in a pipe line connecting the pilot pump **15** and the pilot port of the control valve in the control valve unit **17**, and is configured such that the flow area of the pipe line can be changed. In the illustrated example, the proportional valve **31** operates according to a control instruction output from the controller **30**. Therefore, the controller **30** can adjust the pilot pressure acting on the pilot port of the control valve by the proportional valve **31** regardless of the operation of the operation device **26** by the operator.

[0072] With this configuration, the controller **30** can operate the hydraulic actuator corresponding to the specific operation device **26** even when the operation of the specific operation device **26** is not performed.

[0073] The control system of the work machine **100** includes a controller **30**, a display device D**1**, an input device D**2**, a speech button KS, an external sound collecting device (an example of a sound collecting device) M**1**, an internal sound collecting device M**2**, an external sound output device SP**1**, an internal sound output device SP**2**, an external volume dial DL**1**, an internal volume dial DL**2**, a switch SW, and a communication device T**1**.

[0074] The controller **30** is configured to output a control instruction to the regulator **13** as needed to change the discharge amount of the main pump **14**.

[0075] The controller **30** may be configured to control, for example, the machine guidance function for guiding the manual operation of the work machine **100** by an operator through the operation device **26**. The controller **30** may be configured to control, for example, the machine control function for automatically supporting the manual operation of the work machine **100** by an operator through the operation device **26**.

[0076] Note that some of the functions of the controller **30** may be implemented by other controllers (control devices). That is, the functions of the controller **30** may be implemented in a manner that they are distributed among a plurality of controllers. For example, the machine guidance function and the machine control function may be implemented by an exclusive-use controller (control device).

[0077] Referring now to FIG. **5**, the inside of the driving room **10** will be described. FIG. **5** is a top view of the inside of the driving room **10**. The work machine **100** includes a driving seat **50**, the operation device **26**, a display device D**1**, and the like, which are arranged inside the driving room **10**. An access door is provided on the left side of the driving seat **50**. The operator can enter the inside of the driving room **10** by opening the access door.

[0078] The driving seat **50** is arranged in the center of the driving room **10** in the top view. The driving seat **50** includes a seat **51** on which the operator sits and a backrest **52**. The driving seat **50** is a reclining seat, and the inclination angle of the backrest **52** is adjustable. A left armrest **53**L is arranged on the left side of the driving seat **50**, and a right armrest **53**R is arranged on the right side. The left armrest **53**L and the right armrest **53**R are rotatably supported by the backrest **52**.

[0079] A left console **54**L is arranged on the left side of the driving seat **50**, and a right console **54**R is arranged on the right side. The left console **54**L and the right console **54**R extend along the longitudinal direction. The driving seat **50** is slidable in the longitudinal direction. The driving seat **50** may be slidable in the longitudinal direction together with the left console **54**L and the right console **54**R.

[0080] The left armrest **53**L is arranged on the left console **54**L. The right armrest **53**R is arranged on the right console **54**R. The left armrest **53**L is arranged so as to cover a part of the left console **54**L in the top view. The right armrest **53**R is arranged so as to cover a part of the right console **54**R in the top view.

[0081] The operation device **26** includes a left operation lever **26**L, a right operation lever **26**R, a left traveling pedal **26**PL, a right traveling pedal **26**PR, a left traveling lever **26**DL, and a right traveling lever **26**DR.

[0082] The left operation lever **26**L is provided at the front of the left console **54**L. Similarly, the right operation lever **26**R is provided at the front of the right console **54**R. An operator seated on the driving seat **50** can operate the left operation lever **26**L while holding the left operation lever **26**L with his left hand, and can operate the right operation lever **26**R while holding the right operation lever **26**R with his right hand. An operator seated on the driving seat **50** can operate the left operation lever **26**L with his left hand to drive the arm cylinder **8** and the turning hydraulic motor **2**A. An operator seated on the driving seat **50** can operate the right operation lever **26**R with his right hand to drive the boom cylinder **7** and the bucket cylinder **9**. The bases of the left operation lever **26**L and the right operation lever **26**R are covered with lever boots **27**.

[0083] The left traveling pedal **26**PL and the right traveling pedal **26**PR are arranged on the floor in front of the driving seat **50**. An operator seated on the driving seat **50** can operate the left traveling pedal **26**PL with his left foot to drive the left traveling hydraulic motor **2**ML. The operator seated on the driving seat **50** can operate the right traveling pedal **26**PR with his right foot to drive the right traveling hydraulic motor **2**MR.

[0084] The left traveling lever **26**DL and the right traveling lever **26**DR are arranged between the left traveling pedal **26**PL and the right traveling pedal **26**PR in the top view. The left traveling lever **26**DL and the right traveling lever **26**DR extend upward from the floor surface in front of the driving seat **50**. The operator seated on the driving seat **50** can drive the left traveling hydraulic motor **2**ML in the same manner as the operation through the left traveling pedal **26**PL by gripping the left traveling lever **26**DL with his left hand. The operator seated on the driving seat **50** can drive the right traveling hydraulic motor **2**MR in the same manner as the operation through the right traveling pedal **26**PR by gripping the right traveling lever **26**DR with his right hand. The left traveling lever **26**DL and the right traveling lever **26**DR are arranged such that the operator can simultaneously operate the left traveling lever **26**DL and the right traveling lever **26**DR with one hand.

[0085] The display device D**1** is arranged in front of the right side of the driving seat **50**. The display device D**1** displays various kinds of image information. The display device D**1** includes a display screen for displaying information such as working conditions or operating conditions of the work machine **100**. An operator seated on the driving seat **50** can perform work with the work machine **100** while confirming various kinds of information displayed on the display device D**1**. The display device D**1** may be provided with an input device D**2**.

[0086] The input device D**2** is provided within reach of the operator seated in the driving room **10**, receives various operation inputs by the operator, and outputs signals corresponding to the operation inputs to the controller **30**. The input device D**2** includes a touch panel mounted on the display of the display device D**1** for displaying various information images, a knob switch provided at the tip of one or more of the plurality of operation levers included in the operation device **26**, or a button switch, a lever, a toggle switch, or a rotary dial provided around the display device D**1**. A signal corresponding to the contents of the operation to the input device D**2** is taken into the controller **30**.

[0087] A gate bar **55** is attached to the front of the front end of the left console **54**L. The gate bar **55** operates in conjunction with the operation state of the gate lock lever GL provided in the left console **54**L. The gate bar **55** is mounted on an inner frame of the left console **54**L in a liftable manner around the axis in the lateral direction of the upper end.

[0088] The gate lock lever GL is a mechanical input operation part for switching between a state in which operation of the work machine **100** by the operation device **26** is possible (operable state) and a state in which operation of the work machine **100** by the operation device **26** is not possible (inoperable state). In the illustrated example, the gate lock lever GL is configured such that an operator can switch between a first operation position to implement the inoperable state and a second operation position to implement the operable state. The controller **30** switches between an operable state and an inoperable state according to the operation state of the gate lock lever GL. In the illustrated example, the controller **30** switches between an operable state and an inoperable state of the work machine **100** by electrically switching between communication and non-communication of the pilot line according to the operation state of the gate lock lever GL.

[0089] Further, when the gate lock lever GL is in the second operation position, the gate bar **55** is in a state in which it is raised forward so as to prevent the operator from passing through the access door (passage prohibition state), as illustrated in FIG. **5**. On the other hand, when the gate lock lever GL is in the first operation position, the gate bar **55** is in a state in which it is housed inside the left console **54**L (passage permission state) so as not to prevent the operator from passing through the access door.

[0090] With this configuration, the operator cannot operate the work machine **100** unless the gate lock lever GL is set to the second operating position and the gate bar **55** is set to the passage prohibition state. Therefore, this configuration can prevent the work machine **100** from moving unintentionally even if the operator inadvertently touches the operation device **26** when getting on and off. Therefore, this configuration can improve the safety of the work machine **100**.

[0091] Further, the work machine **100** may be configured to accept a predetermined operation for starting the engine **11** only when the gate lock lever GL is set to the second operating position and the gate bar **55** is set to the passage prohibition state. That is, the work machine **100** may be configured not to start the engine **11** when the gate lock lever GL is set to the first operating position and the gate bar **55** is set to the passage permission state.

[0092] A switch SW is installed in the right console **54**R. A window side console **56** is installed on the right side of the right console **54**R. The window side console **56** extends over the entire length in the longitudinal direction of the driving room **10** and is provided in parallel with the right console **54**R. The display device D**1** is installed at the front of the window side console **56**. The window side console **56** is installed with an external volume dial DL**1**, an internal volume dial DL**2**, and an internal sound collecting device M**2**.
[0093] Further, the window side console **56** is installed with a radio tuner **57** or the like. The radio tuner **57** or the like may be installed in the left console **54**L or the right console **54**R.
[0094] The radio tuner **57** is provided with a switch. On/off control of the radio and adjustment control of the volume can be performed in response to pressing of the switch and rotation operation of the switch. Further, the operation for controlling the radio tuner **57** is not limited to the operation of the physically provided switch, but may be performed by using a touch panel (of the input device D**2**) provided in the display device D**1**. For example, the controller **30** may perform on/off control of the radio tuner **57** and adjustment control of the volume by accepting the operation of the touch panel. The volume adjustment of the radio tuner **57** may be made independent of the volume adjustment of the external voice (the voice of the worker output in the vehicle) or may be made integrally adjustable.
[0095] The internal sound collecting device M**2** is a device for collecting sounds generated in the driving room **10**. In the illustrated example, the internal sound collecting device M**2** is an indoor microphone and is configured to pick up voices emitted by the operator in the driving room **10**.
[0096] The horn button HS is a button operated by the operator of the work machine **100** when the horn is sounded. In the illustrated example, the horn button HS is a knob switch provided at the tip of the left operation lever **26**L.
[0097] The speech button KS is a button operated by the operator of the work machine **100** when the operator of the work machine **100** speaks to a worker around the work machine **100**. In the illustrated example, the speech button KS is a knob switch provided at the tip of the right operation lever **26**R.
[0098] The internal sound output device SP**2** is a device for outputting sound to an operator in the driving room **10** and is provided in the driving room **10**. The internal sound output device SP**2** converts an electric signal input from the controller **30** into a physical sound (vibration of air) and outputs the sound. The internal sound output device SP**2** may be provided at any position, for example, near the display device D**1**, near the input device D**2**, or near the door of the driving room **10**. In the illustrated example, the internal sound output device SP**2** includes a left indoor speaker SP**2**L attached to the upper left corner of the rear wall of the driving room **10** and a right indoor speaker SP**2**R attached to the upper right corner of the rear wall of the driving room **10**. The internal sound output device SP**2** may be a headphone or earphone worn by an operator. In this case, the headphone or earphone is connected so as to be able to communicate with the controller **30** via, for example, Bluetooth (registered trademark).
[0099] The external volume dial DL**1** is configured to adjust the volume of sound output by the external sound output device SP**1**. Further, the volume of sound output by each of the external sound output devices SP**1** may be additionally configured to make adjustments by using a device other than the external volume dial DL**1**, such as a touch panel attached to the display device D**1**.
[0100] The external volume dial DL**1** may be configured to rotate indefinitely in the clockwise and counterclockwise directions. This is to address a case where volume adjustment using the external volume dial DL**1** and volume adjustment using a device other than the external volume dial DL**1** are used together.
[0101] The internal volume dial DL**2** is configured to adjust the volume of sound output by the internal sound output device SP**2**. Further, the volume of sound output by each of the internal sound output devices SP**2** may be additionally configured to make adjustments by using a device other than the internal volume dial DL**2**, such as a touch panel attached to the display device D**1**.

[0102] The internal volume dial DL**2** may be configured to rotate infinitely in both clockwise and counterclockwise directions. This is to address a case where volume adjustment using the internal volume dial DL**2** and volume adjustment using a device other than the internal volume dial DL**2** are used together.
[0103] The storage device **35** is provided, for example, in the driving room **10** and stores various kinds of information under the control of the controller **30**. The storage device **35** is, for example, a nonvolatile storage medium such as a semiconductor memory. The storage device **35** stores information for preventing the output of sound from the internal sound output device SP**2**. The storage device **35** may store, for example, data related to the target work surface acquired through the communication device T**1** or set through the input device D**2**. The target work surface may be set (stored) by the operator of the work machine **100** or set by a work manager or the like.
[0104] The switch SW is a switch for switching whether or not to operate the sound output function for outputting the sound collected by the external sound collecting device M**1** from the internal sound output device SP**2**. In the illustrated example, the switch SW is provided on the upper surface of the right console **54**R. However, the switch SW may be one of the input devices D**2**, may be implemented by a touch panel provided on the display device D**1**, or may be a knob switch.
[0105] A conversation function is a function for implementing a conversation between an operator OP of the work machine **100** and a worker WK around the work machine **100** as illustrated in FIG. **6**. FIG. **6** is a perspective view of the work machine **100** in which the operator OP is seated and the worker WK on the left front side of the work machine **100**. FIG. **6** illustrates how the voice of the operator OP is collected by the internal sound collecting device M**2** and output from the external sound output device SP**1**, and the voice of the worker WK is collected by the external sound collecting device M**1** and output from the internal sound output device SP**2**. In the work machine **100** illustrated in FIG. **6**, the information transmission device G**1** is provided at the top of each of the four sides of the driving room **10**. The front light bar G**1**F provided at the top of the front face of the driving room **10** emits green light, and the left light bar G**1**L provided at the top of the left face of the driving room **10** emits white light. In FIG. **6**, a dot pattern is attached to the front light bar G**1**F that emits green light. When the worker WK sees the front light bar G**1**F that emits green light, the worker WK can recognize that his/her voice is detected by the front microphone M**1**F. In FIG. **6**, other devices such as the imaging device S**6** are not illustrated for clarity.
[0106] The operating state of the conversation function includes an on state (state illustrated in FIG. **6**) in which a conversation between the operator OP and the worker WK is possible, and an off state in which a conversation between the operator OP and the worker WK is not possible. However, the operating state of the conversation function may additionally include at least one of an audible state (relating to the operator OP) in which the operator OP can hear the voice of the worker WK but the worker WK cannot hear the voice of the operator OP and a speech-enabled state (relating to the operator OP) in which the worker WK can hear the voice of the operator OP but the operator OP cannot hear the voice of the worker WK.
[0107] Specifically, when the operating state of the conversation function is switched to the ON state by operating the switch SW, the external sound collecting device M**1**, the external sound output device SP**1**, the internal sound collecting device M**2**, and the internal sound output device SP**2** become available for use. On the other hand, when the operating state of the conversation function is switched to the OFF state by operating the switch SW, the external sound collecting device M**1**, the external sound output device SP**1**, the internal sound collecting device M**2**, and the internal sound output device SP**2** become unavailable for use. When the operating state of the conversation function is switched to the audible state by operating the switch SW, the external sound collecting device M**1** and the internal sound output device SP**2** become available for use. When the operating state of the conversation function is switched to the speech-enabled state by operating the switch SW, the external sound output device SP**1** and the internal sound collecting device M**2** become available for use. In the illustrated example, the operator OP can speak to the

worker WK using the external sound output device SP**1** by speaking while pressing the speech button KS while the internal sound collecting device M**2** is available for use.

[0108] Under the control of the controller **30** according to the present embodiment, the internal sound output device SP**2** provided inside the driving room **10** outputs the sound collected by the external sound collecting device M**1**, and the external sound output device SP**1** provided outside the driving room **10** outputs the sound collected by the internal sound collecting device M**2**. The controller **30** can switch between the output of the sound from the internal sound output device SP**2** and the output of the sound from the external sound output device SP**1** according to the operation of the operator OP, thereby implementing a two-way conversation. The controller **30** according to the present embodiment can prevent the simultaneous output of sound from the internal sound output device SP**2** and the external sound output device SP**1** by performing switching control of sound output according to the operation, and can prevent the occurrence of howling or the like. In the present embodiment, the method of performing the switching control is not limited, and the output of sound from the external sound output device SP**1** and the internal sound output device SP**2** may be performed simultaneously.

[0109] When the controller **30** simultaneously outputs the sound collected by the external sound collecting device M**1** from the internal sound output device SP**2** provided inside the driving room **10** and outputs the sound collected by the internal sound collecting device M**2** from the external sound output device SP**1** provided outside the driving room **10**, the sound output from the sound output devices SP**1** and SP**2** may be collected by the sound collecting devices M**2** and M**1**, resulting in the occurrence of echoes (reverberation) like an "echo among the hills". Therefore, the controller **30** may execute an echo cancellation function for the sound indicated by the sound signal obtained from the external sound collecting device M**1** or the internal sound collecting device M**2**. The echo cancellation function may be a function of eliminating echoes, for example, an echo suppressor system may be used to prevent one person's voice from being picked up while the other person's voice is being heard, or an echo canceller system may be used to eliminate echoes (reverberation) collected by the sound collecting devices M**2** and M**1**. Further, any method may be used for the echo cancellation function, regardless of the well-known method described above.

[Functional Configuration of Controller]

[0110] A configuration for the controller **30** to perform control based on the detected person will be described below. The controller **30** includes an acquiring part **301**, a detecting part **302**, a position estimating part **303**, a sound processing part **304**, an identifying part **305**, a determining part **306**, and an output control part **307** as machine guidance functions and machine control functions.

[0111] The acquiring part **301** acquires detection results from various sensors provided in the work machine **100**. For example, the acquiring part **301** acquires image information indicating an imaging result from the imaging device S**6**. The acquiring part **301** also acquires sound signals representing sounds generated around the work machine **100** from the external sound collecting device M**1**.

[0112] The detecting part **302** performs detection processing of a person existing around the work machine **100** from the image information acquired by the acquiring part **301**. Any method may be used for the detection processing of a person, regardless of the well-known method. For example, it may be determined whether or not the feature quantity extracted from the image information approximates the feature quantity indicating a person by a predetermined value or more.

[0113] When a person is detected by the detecting part **302**, the position estimating part **303** estimates the position of the person in the real space, for example, the direction and distance where the person exists with reference to the work machine **100**. For example, the position estimating part **303** may estimate the direction and distance where the person exists based on the position and size of the person in the image information. Further, when a LiDAR, a distance image camera, or a millimeter wave radar is mounted as the object detection device other than the imaging device S**6**, the direction and distance where the person exists may be estimated based on the detection result obtained by the object detection device. The position estimating part **303** according to the present embodiment acquires the direction and distance where the person exists with reference to the work machine **100** as the position information of the person around the work machine **100**. However, the position information of the person is not limited to the direction and distance where the person exists with reference to the work machine **100**, and information indicating the position coordinates of the person in the world coordinate system may be acquired, for example.

[0114] In the present embodiment, the detection of the person and the estimation of the position of the person are not limited to the above-described method, and the detection of the person and the estimation of the position (for example, the direction and distance) of the person may be performed by using a learned model.

[0115] The sound processing part **304** performs an emphasizing process on the band of the person's voice with respect to the sound signal acquired by the acquiring part **301**, and performs a reduction process on the band other than the band of the person's voice. As the processing, a well-known method may be used, and for example, a noise canceling process may be performed, or the band other than the band of the person's voice may be prevented by a band bus filter or the like. Specifically, as the noise canceling process, the sound processing part **304** performs a process of preventing the sound generated from the work machine **100** such as the engine **11** based on the sound signals from the plurality of external sound collecting devices M**1**. Because the sound generated from the work machine **100** is prevented, the operator can easily hear the sound generated from the environment around the work machine **100**. Because the operator can easily recognize the situation around the work machine **100**, safety can be improved.

[0116] Based on the sound signal processed by the sound processing part **304**, the identifying part **305** identifies whether or not a person who has spoken exists. For example, the identifying part **305** determines whether or not the sound signal from which the sound band of a person's voice is extracted has a volume greater than or equal to a predetermined threshold, and from the determination result, identifies whether or not a person who has spoken exists.

[0117] Further, when the identifying part **305** identifies that a person who has spoken exists, the identifying part **305** identifies the direction of speech based on the volume of each of the sound signals of the plurality of external sound collecting devices M**1**. The direction of speech is, for example, the direction in which a person who has spoken exists. For example, the identifying part **305** identifies, as the direction of speech, the direction in which the external sound collecting device M**1**, which has detected the sound with the highest volume, is located, among the four directions of front, rear, left, and right. Note that the present embodiment indicates an example of the direction identifying method, and is not limited to this method. For example, there is a method for identifying the direction of speech based on the phase shift and the difference in volume of sound signals of a plurality of external sound collecting devices M**1**.

[0118] The determining part **306** determines whether or not the direction of speech identified by the identifying part **305** and the direction in which the person exists estimated by the position estimating part **303** match.

[0119] The output control part **307** outputs information to one or more of the display device D**1** and the internal sound output device SP**2**. For example, when a person who exists around the work machine **100** is speaking, the output control part **307** is configured to report the direction in which the sound caused by the person's speech is detected based on the direction of the sound detected by the external sound collecting devices M**1**. This report is performed when the direction of the person detected from the image information corresponds to the direction of speech.

[0120] For example, the output control part **307** reports the detected direction when the determining part **306** determines that the directions match, and prevents the reporting of the detected direction when the determining part **306** determines that the direction do not match (are different).

[0121] Next, with reference to FIG. **7**, an example of a display screen **42** output by the output control part **307** to the display device D**1** will be described. FIG. **7** is a diagram illustrating an example of a display screen **42** displayed by the display device D**1** according to the present embodiment.

[0122] The output control part **307** according to the present embodiment generates a display screen **42** based on image information input from the imaging device S**6**, sound signals input from the external sound collecting device M**1**, and various kinds of information detected by various sensors or the like provided in the work machine **100**. The various kinds of information detected by the various sensors or the like include a turning angle, information indicating an area in which a person is photographed in the image information, position information of a person present around the work machine **100** (information indicating the direction and distance in which the detected person is present), a person who has spoken based on sound signals input from the external sound collecting device M**1**, and detection results of various sensors.

[0123] The display screen **42** includes a date and time display area **42***a*, a travel mode display area **42***b*, an attachment display area **42***c*, a fuel consumption display area **42***d*, an engine control state display area **42***e*, a cooling water temperature display area **42***g*, a fuel remaining amount display area **42***h*, a rotation speed level display area **42***i*, a urea water remaining amount display area **42***j*, a hydraulic oil temperature display area **42***k*, a state display area **1421**, a first image display area **1422**, a second image display area **1423**, and a third image display area **1424**, according to the control from the output control part **307**. The display screen **42** may include other display areas.

[0124] The travel mode display area **42***b*, the attachment display area **42***c*, the engine control state display area **42***e*, and the rotation speed level display area **42***i* are areas for displaying setting state information, which is information related to the setting state of the work machine **100**. The fuel consumption display area **42***d*, the engine operation time display area, the cooling water temperature display area **42***g*, the fuel remaining amount display area **42***h*, the urea water remaining amount display area **42***j*, and the hydraulic oil temperature display area **42***k* are areas for displaying operation state information, which is information representing the operation state of the work machine **100**, based on the detection results of various sensors.

[0125] The date and time display area **42***a* is an area for displaying the current date and time. The travel mode display area **42***b* is an area for displaying the current travel mode. The attachment display area **42***c* is an area for displaying an image representing the currently attached attachment. The fuel consumption display area **42***d* is an area for displaying fuel consumption information calculated by the controller **30**. The fuel consumption display area **42***d* includes an average fuel consumption display area **42***d***1** for displaying lifetime average fuel consumption or section average fuel consumption and an instantaneous fuel consumption display area **42***d***2** for displaying instantaneous fuel consumption.

[0126] The engine control state display area **42***e* is an area for displaying the control state of the engine **11**. The engine operation time display area is an area for displaying the cumulative operation time of the engine **11**. The cooling water temperature display area **42***g* displays the current temperature state of the engine cooling water. The fuel remaining amount display area **42***h* displays the remaining amount of fuel stored in the fuel tank.

[0127] The rotation speed level display area **42***i* displays the current level set by the dial (not illustrated) in an image. A number indicating the selected level is displayed in the rotation speed level display area **42***i*. The number “1” displayed in the rotation speed level display area **42***i* indicates that the selected rotation speed level is the “1st level”. The number “n” displayed in the rotation speed level display area **42***i* indicates that the selected rotation speed level is the “nth level”. The notation of “n” is a natural number. When the operator rotates the dial (not illustrated), the number displayed in the rotation speed level display area **42***i* changes.

[0128] The urea water remaining amount display area **42***j* is an area for displaying the remaining amount of urea water stored in the urea water tank with an image. The hydraulic oil temperature display area **42***k* is an area for displaying the temperature state of the hydraulic oil in the hydraulic oil tank.

[0129] In the screen illustrated in FIG. **7**, the first image display area **1422**, the second image display area **1423**, and the third image display area **1424** are areas for displaying image information captured by the imaging device S**6**. The first image display area **1422** displays a right image. The second image display area **1423** displays a rear image. The third image display area **1424** displays a left image. The right image is an image reflecting the space on the right side of the work machine **100**, the rear image is an image reflecting the space behind the work machine **100**, and the left image is an image reflecting the space on the left side of the work machine **100**.

[0130] The first image display area **1422** is displayed right with reference to the state display area **1421**. The second image display area **1423** is displayed downward with reference to the state display area **1421**. The third image display area **1424** is displayed left with reference to the state display area **1421**. It is assumed that the lower portion of the display screen **42** corresponds to the rear of the work machine **100**. That is, the display screen **42** displays image information captured by the imaging device S**6** in the direction captured by the imaging device S**6** with the state display area **1421** as a reference. The present embodiment describes an example of the arrangement of image information, and is not limited to this arrangement. For example, the image display area may be arranged regardless of the direction captured. In the present embodiment, because the image information captured is displayed in the direction captured with the state display area **1421** as a reference, the operator can intuitively recognize which direction the image information represents when referring to the image information. Therefore, safety can be improved.

[0131] When the position and distance of the person are identified by the position estimating part **303**, the output control part **307** superimposes a frame indicating that the person has been detected in the area identified by the identified position and distance on one or more of the right image, rear image, and left image.

[0132] As a result, a frame **1422***b* is displayed on the right image of the first image display area **1422** so as to surround the person **1422***a*, and a frame **1424***b* is displayed on the left image of the third image display area **1424** so as to surround the person **1424***a*.

[0133] The state display area **1421** is an area for displaying information representing a positional relationship between the work machine **100** and a person detected around the work machine **100**.

[0134] The state display area **1421** is a display area representing a real space around the work machine **100** at a predetermined scale. In the state display area **1421**, a work machine icon (an example of display information) **1421***b* indicating the presence of the work machine **100** is arranged at the center of the area.

[0135] Other than the work machine icon **1421***b* representing the work machine **100**, a direction display icon **1421***a* indicating a moving direction when the work machine **100** advances and an icon (in the example of FIG. **7**, the person detection icons **1421***e*, **1421***f*, and **1421***g*) representing a person detected around the work machine **100** are simultaneously displayed in the state display area **1421**. In the state display area **1421**, areas (that is, the background) other than the work machine icon **1421***b*, the direction display icon **1421***a*, the person detection icon **1421***e*, **1421***f*, **1421***g*, and the area **1421***h* may be areas represented by, for example, a single color (e.g., black).

[0136] The work machine icon **1421***b* is an icon that combines an image indicating the upper turning body **3** and an image indicating the lower traveling body **1** according to the positional relationship between the upper turning body **3** and the lower traveling body **1** based on the turning angle.

[0137] The direction display icon **1421***a* indicates, in a triangular shape, the direction in which the work machine **100** travels when the traveling lever is tilted forward. Note that the present embodiment describes an example of an icon indicating the direction in which the work machine **100** travels when the traveling lever is tilted forward, and any shape may be used as long as it indicates the direction in which the work machine **100** can travel.

[0138] The person detection icons **1421***e*, **1421***f*, and **1421***g* are icons representing persons detected by the image information captured by the imaging device S**6**. Specifically, the person detection icons **1421***e*, **1421***f*, and **1421***g* are arranged based on the estimation result of the position and direction of a person estimated by the position estimating part **303**. For example, the person detection icons **1421***e*, **1421***f*, and **1421***g* are arranged at positions obtained by multiplying the direction and distance of a person indicated by the estimation result by a predetermined scale rate with the work machine **100** as a reference.

[0139] As described above, the positional relationship between the work machine icon **1421***b* and the person detection icons **1421***e*, **1421***f*, and **1421***g* corresponds to the positional relationship between the work machine **100** in the real space and the person existing around the work machine **100**.

[0140] Further, the operator can estimate how the positional relationship between the work machine **100** and a person existing in the vicinity changes when the work machine **100** is moved, by referring to the state display area **1421**.

[0141] Further, the state display area **1421** displays a first circular area **1421***c* and a second circular area **1421***d* determined according to the distance from the work machine **100** with the work machine **100** as a reference.

[0142] The first circular area **1421***c* is information indicating the turning range of the current attachment AT of the work machine **100** based on the detection results obtained by the boom angle sensor S**1**, the arm angle sensor S**2**, and the bucket angle sensor S**3**.

[0143] The second circular area **1421***d* is information indicating the turning range of the attachment AT when the attachment AT is extended most in the horizontal direction.

[0144] The display screen **42** according to the present embodiment displays a first circular area **1421***c* and a second circular area **1421***d*. That is, the positional relationship between a person and the attachment AT can be recognized by the positional relationship between the first circular area **1421***c*, the second circular area **1421***d*, and the person detection icons **1421***e*, **1421***f*, and **1421***g*. Therefore, the operator can easily perform an operation in which the person does not come into contact with the attachment AT, thereby improving safety.

[0145] Further, the output control part **307** changes the display mode of the person detection icons **1421***e*, **1421***f*, and **1421***g* according to whether they are included in the first circular area **1421***c* and the second circular area **1421***d*.

[0146] For example, the person detection icon **1421***e* existing in the first circular area **1421***c* is displayed in red. The person detection icon **1421***f* existing outside the first circular area **1421***c* and in the second circular area **1421***d* is displayed in yellow, for example. The person detection icon **1421***g* existing outside the second circular area **1421***d* is displayed in blue, for example. The output control part **307** according to the present embodiment implements different display modes depending on whether or not the icon is included in the first circular area **1421***c* and the second circular area **1421***d*. That is, the color of the person detection icon changes according to the distance from the work machine **100**. In this way, the display device D**1** can alert the operator according to the distance between the work machine **100** and the person by displaying the person detection icon whose color is changed according to the distance. Therefore, safety can be improved.

[0147] By referring to the state display area **1421**, the operator can recognize how the positional relationship between the attachment AT of the work machine **100** and the person existing in the vicinity changes when the work machine **100** is turned. Therefore, safety can be improved.

[0148] The output control part **307** matches the color of the frame **1422***b* of the right image of the first image display area **1422** with the color of the person detection icon **1421***e*, and matches the color of the frame **1424***b* of the rear image of the third image display area **1424** with the color of the person detection icon **1421***f*.

[0149] That is, a frame indicating the detected person is displayed on the display screen **42** for the right image and the left image, and the correspondence relationship between the person indicated by the frame and the person detection icon is displayed recognizable. Thus, the operator can recognize the situation of the person indicated by the state display area **1421** by referring to the right image and the left image.

[0150] Further, the output control part **307** displays an area **1421***h*, which is identified by the identifying part **305** and indicates the direction of speech, on the state display area **1421**. The area **1421***h* of the state display area **1421** is expressed by a color different from that of the background (another area) of the state display area **1421**. That is, the output control part **307** according to the present embodiment displays the area **1421***h* (an example of information), which indicates the direction in which the sound from the person's speech was detected, on the display device D**1** in a manner different from that of the another area (an example of information) with reference to the work machine icon **1421***b*. Thus, the operator can recognize the direction of speech by referring to the state display area **1421**. Then, the operator can estimate which person spoke from the direction. For example, the operator can estimate that the person corresponding to the person detection icon **1421***f* included in the area **1421***h* spoke to the operator. Therefore, because the operator can estimate the person who spoke to the operator, it is easy to communicate with the person who spoke to the operator. This reduces the burden at the time of work.

[0151] Further, there is a case where the direction of speech identified by the identifying part **305** and the direction in which a person exists identified by the position estimating part **303** are slightly different.

[0152] FIGS. **8**A and **8**B are diagrams explaining an example of correcting the direction of speech in the output control part **307** according to the present embodiment. FIG. **8**A illustrates an example in which the identifying part **305** detects a sound caused by a person's speech from the left of the four directions. In this case, the output control part **307** expresses the area **1502** corresponding to the left direction with a color different from that of other areas. However, in FIGS. **8**A and **8**B, the position estimating part **303** estimates that a person exists in the direction and distance indicated by the person detection icon **1501**.

[0153] The direction of speech identified by the identifying part **305** may be shifted depending on the surrounding environment. The output control part **307** according to the present embodiment corrects the area indicating the direction of speech according to the position where a person actually exists.

[0154] The output control part **307** corrects the area indicating the direction of speech based on the direction of the person detected from the image information identified by the position estimating part **303**. The reference to be corrected may be any reference. For example, even if the direction of the person estimated by the position estimating part **303** is not included in the area indicating the direction identified by the identifying part **305**, the output control part **307** corrects the area so that the direction of the person estimated by the position estimating part **303** is included if it is closer than a predetermined reference (for example, in the case of less than or equal to 10 degrees).

[0155] FIG. **8**B illustrates the state display area **1421** after the output control part **307** has corrected the area. The area **1503** illustrated in FIG. **8**B has been corrected to include the person detection icon **1421***f*. Therefore, by referring to the state display area **1421** illustrated in FIG. **8**B, the operator can recognize the person who has spoken to him/her and communicate with the person. As described above, in the present embodiment, even if the direction of speech identified by the external sound collecting device M**1** is shifted, the output control part **307** corrects the direction of speech based on the direction of the person detected from the image information, so that the direction of speech can be presented to the operator without giving the feeling that something is wrong. Therefore, when the operation of the work machine **100** is required based on the spoken content, the operator can identify which direction the spoken content is based and which direction the work machine **100** is to be operated, such that convenience can be improved.

[0156] FIG. **7** illustrates an example of the display screen **42** displayed by the output control part **307**, but the direction of speech may be expressed by another mode.

[0157] FIG. **9** illustrates an example of a display screen **42**A displayed by the display device D**1** according to the present embodiment. The same display as in FIG. **7** is assigned the same reference numerals and a description thereof is omitted.
[0158] In the screen illustrated in FIG. **9**, the first image display area **1422**, the second image display area **1423**, and the third image display area **1424** are areas for displaying image information captured by the imaging device S**6**.
[0159] Further, the output control part **307** displays information indicating the direction of speech identified by the identifying part **305** on the outer frame of one or more of the first image display area **1422**, the second image display area **1423**, and the third image display area **1424**.
[0160] In the example illustrated in FIG. **9**, the outer frame **1424**FL of the third image display area **1424** is thicker than the outer frame **1422**FL of the first image display area **1422** and the outer frame **1423**FL of the second image display area **1423**. Further, the outer frame **1424**FL is expressed by a different color to that of the outer frames **1422**FL and **1422**FL.
[0161] The operator can recognize the direction of speech by referring to the first image display area **1422**, the second image display area **1423**, and the third image display area **1424**. That is, because the outer frame **1424**FL has a different display mode from that of the other outer frames, the operator can presume that the person who appears in the third image display area **1424** surrounded by the outer frame **1424**FL has spoken. Therefore, because the operator can estimate the person who has spoken, it becomes easy to communicate with the person who has spoken.
[0162] If the direction of speech is the forward direction, the person who has spoken does not appear in any of the first image display area **1422**, the second image display area **1423**, or the third image display area **1424**. In this case, there is a method in which the output control part **307** displays a message on the display device D**1** indicating that a voice is heard from the front side and the screen is switched to a screen in which the front side is appearing. As another method, there is a method in which the output control part **307** automatically switches to a display screen including an image which is captured by the camera S**6**F and in which the space in front of the work machine **100** is appearing.
[0163] In the example illustrated in FIG. **9**, the output control part **307** displays the image information captured by the imaging device S**6** on the display device D**1**, and displays the direction in which the sound caused by a person's speech was detected by changing the display mode of the outer frame of the captured image information. Therefore, the operator can recognize the direction of the speech and the current situation of the direction. Therefore, the operator can easily estimate the person who spoke and communicate with the person who spoke.
[0164] The display screens **42** and **42**A illustrated in FIGS. **7** to **9** illustrate one mode of displaying the surrounding area of the work machine **100**, and the surrounding area of the work machine **100** may be expressed by other modes. As a method of displaying the surrounding area of the work machine **100**, a bird's-eye view image may be used.
[0165] FIG. **10** illustrates an example of a display screen **42**B displayed by the display device D**1** according to the present embodiment. The same display as that illustrated in FIG. **7** is assigned the same reference numerals and a description thereof is omitted.
[0166] In the screen illustrated in FIG. **10**, a bird's-eye view image **1700** is image information expressing the surrounding area of the work machine **100** from above the work machine **100** with reference to the work machine icon **1701**. The bird's-eye view image **1700** is image information generated by each of the four cameras S**6**F, S**6**B, S**6**L, and S**6**R based on image information. The output control part **307** according to the present embodiment combines the image information captured from the front, the image information captured from the rear, the image information captured from the right, and the image information captured from the left, and converts them into an image seen from just above the work machine **100** to generate the bird's-eye view image **1700**.
[0167] In the example illustrated in FIG. **10**, the output control part **307** superimposes a thick line **1704** indicating the area where the person who spoke is present on the outer frame **1702** of the

bird's-eye view image **1700** to be displayed. The thick line **1704** may have a different color than that of the outer frame **1702** of the bird's-eye view image **1700**.

[0168] By referring to the bird's-eye view image **1700**, the operator can recognize the left direction indicated by the thick line **1704** as the direction of speech with reference to the work machine **100**. Therefore, the operator can assume that the person **1703** in the bird's-eye view image **1700** has spoken to the operator.

[0169] Therefore, because the controller **30** according to the present embodiment can estimate the person who spoke, it is easy to communicate with the person who spoke to the operator. Therefore, the work load is reduced.

[0170] Although the example illustrated in FIG. **10** illustrates an example of displaying only the bird's-eye view image **1700**, the mode is not limited to the mode of displaying only the bird's-eye view image **1700**, and the bird's-eye view image **1700** may be displayed in combination with any one or more of the first image display area **1422** of the right image, the second image display area **1423** of the rear image, and the third image display area **1424** of the left image as illustrated in FIGS. **8**A to **9**. For example, there is a method in which the output control part **307** outputs a display screen in which the bird's-eye view image **1700**, the first image display area **1422**, and the second image display area **1423** are combined.

[0171] Next, the processing procedure executed by the controller **30** according to the present embodiment will be described. FIG. **11** is a flowchart illustrating the processing procedure for displaying the direction of speech by the controller **30** according to the present embodiment.

[0172] First, the acquiring part **301** acquires image information obtained by capturing the surrounding area of the work machine **100**, from the imaging device S**6** (S**1801**).

[0173] The detecting part **302** detects a person existing around the work machine **100** from the image information acquired by the acquiring part **301** (S**1802**).

[0174] When a person is detected by the detecting part **302**, the position estimating part **303** estimates the direction and distance in which the person exists with reference to the work machine **100** (S**1803**).

[0175] Next, the acquiring part **301** acquires sound signals representing sounds generated around the work machine **100** from each of the plurality of external sound collecting devices M**1** (S**1804**).

[0176] With respect to the sound signals of each of the external sound collecting devices M**1** acquired by the acquiring part **301**, the sound processing part **304** performs an emphasizing process on the band of the person's voice and performs a reduction process (noise removal process) for the band other than the band of the person's voice (S**1805**).

[0177] The identifying part **305** identifies the direction of speech based on the sound signal processed by the sound processing part **304** (S**1806**).

[0178] The determining part **306** determines whether the direction of speech identified by the identifying part **305** corresponds to the direction in which a person exists estimated by the position estimating part **303** (S**1807**).

[0179] If the determining part **306** determines that the directions of speech do not match (S**1807**: NO), the display of the direction of speech is prevented and the process proceeds to S**1809**.

[0180] On the other hand, if the determining part **306** determines that the directions of speech match (S**1807**: YES), the output control part **307** displays the direction of speech (S**1808**). For example, the output control part **307** performs display in any one of the modes of the display screen **42** in FIG. **7**, the display screen **42**A in FIG. **9**, and the display screen **42**B in FIG. **10**.

[0181] Further, the output control part **307** outputs the sound signal processed by the sound processing part **304** (S**1809**).

[0182] In the present embodiment, the operator can recognize the direction of speech around the work machine **100** by performing the processing procedure described above. Because the operator can estimate the person who spoke by visually recognizing the direction of speech, it is easy to communicate with the person who spoke.

[0183] Further, in the present embodiment, by performing an emphasizing process on the band of the person's voice and outputting the sound signal from which noise is removed, it is easy for the operator to identify what the person who exists around the work machine **100** said.
[0184] The present embodiment describes an example of the arrangement of the external sound collecting device M**1**, but the arrangement is not limited to this arrangement example. For example, one or more microphone arrays may be provided in the upper turning body **3**. The microphone array has a plurality of microphones, and the direction in which the sound is generated can be recognized by processing and analyzing the difference (time difference, phase difference) of the sound perceived by the plurality of microphones. When one microphone array having such directivity is provided, the direction of the sound can be detected from the sound detected by each of the microphones included in the microphone array.
[0185] In the present embodiment, an example of providing four external sound collecting devices M**1** has been described. However, the present embodiment does not limit the number of external sound collecting devices M**1** attached to the work machine **100**, and the number may be three or less, or five or more.
Modified Example
[0186] In the present embodiment, an example of displaying the direction on the display device D**1** has been described as the reporting method. However, the above-described embodiment does not limit the reporting method to the method of displaying the direction on the display device D**1**. In this modified example, a plurality of internal sound output devices SP**2** are provided in the driving room **10**. The internal sound output devices SP**2** are provided, for example, in the driving room **10** at the front, rear, left, and right directions with reference to the operator seat.
[0187] The identifying part **305** according to this modified example identifies the direction of speech based on sound signals from the four external sound collecting devices M**1**, and the output control part **307** controls the output of sound from the internal sound output device SP**2** corresponding to the direction of speech among the plurality of internal sound output devices SP**2**. In this modified example, the operator can recognize the direction of speech from the direction in which the sound is output. Note that the work machine **100** according to this modified example does not have to be provided with the imaging device S**6**.
[0188] This modified example describes one mode of the reporting method, and other modes may be used. For example, the driving room **10** may be provided with a plurality of light sources, and the output control part **307** may control the light source corresponding to the direction of speech among the plurality of light sources. As another example, each of the plurality of operation levers of the operation device **26** may be provided with a vibration mechanism, and the output control part **307** may control the vibration mechanism corresponding to the direction existing in the direction of speech to vibrate, among the plurality of operation levers. In this modified example, because the right and left operation levers are provided, the direction of speech can be intuitively recognized only in the lateral direction. In this modified example, the operator can recognize the direction of speech by performing the above-described reporting method. Therefore, the operator can estimate the person who spoke by visually recognizing the direction of speech, and it becomes easy to communicate with the person who spoke.
[0189] The reporting method is not limited to one mode, and a plurality of modes may be combined. For example, two or more of the following may be combined: display of the direction on the display device D**1**, output of sound from the internal sound output device SP**2** corresponding to the direction, lighting of the light source corresponding to the direction, and vibration of each of the plurality of operation levers of the operation device **26**.
Second Embodiment
[0190] Next, another configuration example of the work machine **100** according to the second embodiment will be described with reference to FIG. **12**. FIG. **12** is a top view of another configuration example of the work machine **100** according to the second embodiment. The work

machine **100** illustrated in FIG. **12** differs from the work machine **100** illustrated in FIG. **1** in that the external sound output device SP**1** includes four speakers (a front speaker SP**1**F, a left speaker SP**1**L, a right speaker SP**1**R, and a rear speaker SP**1**B). In the work machine **100** illustrated in FIG. **1**, the external sound output device SP**1** includes one non-directional speaker provided above the driving room **10**.

[0191] With this configuration, the work machine **100** illustrated in FIG. **12** can output sound toward the workers WK in front of the work machine **100** without outputting sound toward the worker WK at the left, right, and rear of the work machine **100**, by turning on the front speaker SP**1**F (the state in which sound can be output) and turning off the left speaker SP**1**L, the right speaker SP**1**R, and the rear speaker SP**1**B (the state in which sound cannot be output), for example.

[0192] In the example illustrated in FIG. **12**, the front camera S**6**F and the front microphone M**1**F are provided adjacent to the front speaker SP**1**F, and the front light bar G**1**F is provided in the case of the front microphone M**1**F. The left camera S**6**L and the left microphone M**1**L are provided adjacent to the left speaker SP**1**L, and the left light bar G**1**L is provided in the case of the left microphone M**1**L. The right camera S**6**R and the right microphone M**1**R are provided adjacent to the right speaker SP**1**R, and the right light bar G**1**R is provided in the case of the right microphone M**1**R. The rear camera S**6**B and the rear microphone M**1**B are provided adjacent to the rear speaker SP**1**B, and the rear light bar G**1**B is provided in the chassis of the rear microphone M**1**B.

[0193] In this case, the work machine **100** may turn on the light bar corresponding to the turned on speaker (light emission enabled state) and turn off the light bar corresponding to the turned off speaker (light emission disabled state).

[0194] The external sound output device SP**1** may include one or a plurality of parametric speakers. The parametric speaker is a speaker using ultrasonic waves that can selectively transmit sound to a person within a specific narrow range. The parametric speaker can transmit sound to any position.

[0195] In the work machine **100** illustrated in FIG. **12**, the controller **30** may detect the worker WK around the work machine **100** based on the image captured by the imaging device S**6**, and identify the position of the worker WK. If there are a plurality of workers WK around the work machine **100**, the controller **30** may discriminate between the conversation target (the worker WK who is speaking) and the non-target (the worker WK who is not speaking) based on the outputs of the four external sound collecting devices M**1**. The controller **30** may discriminate between the conversation target (the worker WK who is facing the work machine **100**) and the non-target (the worker WK who is not facing the work machine **100**) based on the image captured by the imaging device S**6**. Then, the controller **30** may turn on the speaker and the light bar facing the worker WK. For example, if there is a worker WK (speaking) behind the work machine **100**, the controller **30** may turn on the rear speaker SP**1**B while keeping the front speaker SP**1**F, the left speaker SP**1**L, and the right speaker SP**1**R off. In this case, the controller **30** may turn on the rear light bar G**1**B while keeping the front light bar G**1**F, the left light bar G**1**L, and the right light bar G**1**R off. Such a function may be implemented in the work machine **100** illustrated in FIGS. **1** to **5**.

[0196] With this configuration, the controller **30** can output sound in the direction where the worker WK is located without outputting sound in the direction where the worker WK is not located. Therefore, the worker WK can easily recognize whether he/she is a conversation target or a non-target.

Third Embodiment

[0197] In the third embodiment, a case where the operator performs remote operation of the work machine **100** will be described.

[0198] Next, with reference to FIG. **13**, a configuration example of the operation system (one example of a control system) SYS according to the third embodiment will be described. FIG. **13** is a schematic diagram illustrating a configuration example of the operation system (one example of a control system of a work machine) SYS of the work machine **100**. As illustrated in FIG. **13**, the operation system SYS includes the work machine **100**, the remote control room RC, and the management center MC. In FIG. **13**, the detailed configuration of the work machine **100** is not illustrated. This is because the work machine **100** illustrated in FIG. **13** has the same configuration as the work machine **100** illustrated in FIG. **1** or **10**.

[0199] The work machine **100**, the remote control room RC, and the management center MC are connected to each other so that data can be transmitted and received via the communication network NW. The work machine **100**, the remote control room RC, and the management center MC may be connected to each other so that data can be transmitted and received directly without the communication network NW. In the illustrated example, the work machine **100** transmits information about the work site to the remote control room RC. Thus, the remote operator RO in the remote control room RC can identify the situation of the work site based on the information from the work machine **100**.

[0200] For example, the work machine **100** transmits image information captured by the imaging device S**6** and a sound signal representing sound collected by the external sound collecting device M**1** to the remote control room RC.

[0201] The work machine **100** is provided with a sensor capable of recognizing the position and shape of objects in the work site in three dimensions. For example, the work machine **100** is provided with a space recognition device. Therefore, the work machine **100** can transmit the result of measuring the work site in three dimensions to the remote control room RC.

[0202] The space recognition device is a device for recognizing the space around the work machine **100**. In the illustrated example, the space recognition device is a LiDAR. The LiDAR measures the distance between each of 1 million or more points within the monitoring range and the LiDAR. The space recognition device may be any device capable of measuring the distance to an object. For example, the space recognition device may be a stereo camera or a combination of the imaging device S**6** and a distance measuring device such as a millimeter wave radar.

[0203] The work machine **100** included in the operation system SYS may be one or more machines. When a plurality of work machines **100** are included, the remote operator RO of a specific work machine **100** can obtain information on the working site obtained by one or more other work machines **100** in addition to the information on the working site obtained by the specific work machine **100**.

[0204] A communication device T**2**, a remote controller R**40**, an operation device R**42**, an operation sensor R**43**, a display device D**1**E, an internal sound collecting device M**2**E, and an internal sound output device SP**2**E are installed in the remote control room RC. An operation seat DS on which the remote operator RO who remotely operates the work machine **100** sits is installed in the remote control room RC.

[0205] The communication device T**2** is configured to communicate with the communication device T**1** attached to the work machine **100**.

[0206] The remote controller R**40** is an operation device that executes various operations. In the present embodiment, the remote controller R**40** is composed of a microcomputer including a CPU and a memory. Various functions of the remote controller R**40** are implemented by the CPU executing programs stored in the memory.

[0207] The display device DiE can display various kinds of information. The display device DiE displays an image based on information transmitted from the work machine **100** in order for the remote operator RO in the remote control room RC to visually recognize the surroundings of the work machine **100**. In the illustrated example, the display device DiE is a liquid crystal display. Note that the display device DiE may be a display or a projector for implementing the naked-eye stereoscopic vision, or VR goggles or the like.

[0208] The internal sound collecting device M**2**E is a device for collecting sounds generated in the remote control room RC. In the illustrated example, the internal sound collecting device M**2**E is an indoor microphone, and is configured to pick up voices emitted by the remote operator RO in the remote control room RC.

[0209] The internal sound output device SP**2**E can output various kinds of sound information. The internal sound output device SP**2**E outputs sound based on information transmitted from the work machine **100** so that the remote operator RO in the remote control room RC can hear the sound emitted at the work site. For example, the internal sound output device SP**2**E may be configured to output the sound captured by the external sound collecting device M**1** mounted outside the driving room **10** or the internal sound collecting device M**2** mounted inside the driving room **10**. In this case, the internal sound collecting device M**2** may be provided in the driving room **10** at a position corresponding to the ear position of the operator sitting on the driving seat **50**. The internal sound output device SP**2**E may be an installed device such as a speaker or a wearable device such as an earphone or a headphone. The speaker may be a monaural speaker, a stereo speaker, or a surround speaker. The speaker may be a non-directional speaker or a directional speaker. The wearable device may have a noise canceling function, a spatial audio function (stereo sound function), or a bone conduction function.
[0210] An operation sensor R**43** for detecting the operation contents of the operation device R**42** is installed in the operation device R**42**. The operation sensor R**43** is, for example, an inclination sensor for detecting the inclination angle of the operation lever or an angle sensor for detecting the oscillation angle of the operation lever around the oscillation shaft. The operation sensor R**43** may be composed of other sensors such as a pressure sensor, a current sensor, a voltage sensor, or a distance sensor. The operation sensor R**43** outputs information about the detected operation contents of the operation device R**42** to the remote controller R**40**. The remote controller R**40** generates an operation signal based on the received information and transmits the generated operation signal to the work machine **100**. The operation sensor R**43** may be configured to generate an operation signal. In this case, the operation sensor R**43** may output the operation signal to the communication device T**2** without passing through the remote controller R**40**. With this configuration, the remote operator RO can remotely operate the work machine **100** from the remote control room RC.
[0211] Like the controller **30** of the first embodiment, the remote controller R**40** includes an acquiring part **301**, a detecting part **302**, a position estimating part **303**, a sound processing part **304**, an identifying part **305**, a determining part **306**, and an output control part **307**.
[0212] The acquiring part **301** of the remote controller R**40** acquires a sound signal representing the sound collected by the external sound collecting device M**1** from the work machine **100** via the communication device T**2**. Further, the acquiring part **301** acquires a sound signal representing the sound collected by the internal sound collecting device M**2**E of the remote control room RC.
[0213] The output control part **307** of the remote controller R**40** can implement control, via the communication device T**2**, to output the sound based on the sound signal acquired from the external sound collecting device M**1** by the acquiring part **301**, from the internal sound output device SP**2**E of the remote control room RC. Thus, the remote operator RO can hear the sound of a person existing around the work machine **100**.
[0214] The output control part **307** of the remote controller R**40** can implement control, via the communication device T**2**, to output the sound based on the sound signal acquired from the internal sound collecting device M**2**E of the remote control room RC by the acquiring part **301**, from the external sound output device SP**1** of the work machine **100**. Thus, a person existing around the work machine **100** can hear the voice of the remote operator RO.
[0215] The detecting part **302**, the position estimating part **303**, the sound processing part **304**, the identifying part **305**, and the determining part **306** of the remote controller R**40** perform the same control as the controller **30** of the above-described embodiment.
[0216] With this configuration, the remote controller R**40**, similar to the controller **30** described above, identifies the direction of speech based on the acquired sound signal. Then, the remote controller R**40** displays the display screen **42** illustrated in FIG. **7**, the display screen **42**A illustrated in FIG. **9**, or the display screen **42**B illustrated in FIG. **10** on the display device DR as in

the embodiment described above.
[0217] The management center MC is a facility provided with various devices for managing the remote operation of the work machine **100** by the remote operator RO in the remote control room RC. In the illustrated example, the management center MC is installed at a position apart from the work site of the work machine **100** and the remote control room RC. In the management center MC, the management device **200**, the internal sound collecting device M**2**C, and the internal sound output device SP**2**C are installed.
[0218] The management device **200** is an example of a control device, such as a server computer (what is referred to as a cloud server) or an edge server. The management device **200** is typically a fixed terminal device, but may be a portable terminal device (for example, a laptop computer, a tablet, or a smartphone). The management device **200** can perform the same control as the remote controller R**40**.
[0219] With this configuration, the manager in the management center MC can listen to the sound generated at the work site by using, for example, the sound collecting device (the external sound collecting device M**1** or the internal sound collecting device M**2**) and the internal sound output device SP**2**C attached to the work machine **100**. The manager in the management center MC can listen to the sound generated in the remote control room RC by using, for example, the internal sound collecting device M**2**E and the internal sound output device SP**2**C attached to the remote control room RC. Further, the manager in the management center MC can transmit his/her voice to the worker WK around the work machine **100** by using, for example, the internal sound collecting device M**2**C and the external sound output device SP**1** attached to the work machine **100**. The manager in the management center MC can transmit his/her voice to the remote operator RO in the remote control room RC by using, for example, the internal sound collecting device M**2**C and the internal sound output device SP**2**E attached to the remote control room RC.
[0220] The management device **200** has the same configuration as the remote controller R**40**. Therefore, the management device **200** displays the display screen **42** illustrated in FIG. **7**, the display screen **42**A illustrated in FIG. **9**, or the display screen **42**B illustrated in FIG. **10** on the display device DR as in the above-described embodiments.
[0221] In the present embodiment, the internal sound collecting device M**2**C and the internal sound output device SP**2**C implement the same functions as the internal sound collecting device M**2**E and the internal sound output device SP**2**E, and the management device **200** transmits and receives sound signals with the work machine **100** or the remote control room RC. Therefore, the manager in the management center MC implements a two-way conversation between the remote operator RO in the remote control room RC and the person (including the worker WK) who exists around the work machine **100**. The specific control for implementing the two-way conversation is the same as in the above-described embodiments, and a description is omitted.
[0222] Thus, in the present embodiment, a two-way conversation is implemented between the remote operator RO on the operation seat DS and the person who exists around the work machine **100**. Similarly, a two-way conversation is implemented between the manager who manages the work site with the management device **200** and the person who exists around the work machine **100**. Therefore, it is easy to communicate with the person who exists around the work machine **100** even in the case of remote operation or remote management.
[0223] Further, in the present embodiment, the remote operator RO on the operation seat DS and the manager in the management center MC can confirm the direction of speech around the work machine **100**. Therefore, even when remote control or remote management is performed, it is easy to communicate with a person who exists around the work machine **100**.
<Functions>
[0224] In the above-described embodiments and modified example, the operator can recognize the direction of speech around the work machine **100**. Therefore, the operator can easily communicate with the person who has spoken. Further, because the operator can identify the situation around the

person who has spoken, the operator can recognize the intention of the speech. The operator can know how to operate the work machine **100** according to the intention of the speech. Therefore, in the embodiments and the modified example, the work efficiency of the work machine **100** can be improved.

[0225] The work machine and the control system of the work machine according to the present invention have been described in the above embodiments, but the present invention is not limited to the above-described embodiments. Various changes, corrections, substitutions, additions, deletions and combinations are possible within the scope of the claims. These also naturally fall within the scope of the present invention. Additionally, the modifications are included in the scope of the invention.

Claims

1. A work machine comprising: a lower traveling body; an upper turning body mounted on the lower traveling body so as to freely turn; a sound collecting device provided on the upper turning body so as to be able to detect a direction of a sound around the work machine; and a control device configured to report a direction of speech based on the direction of the sound detected by the sound collecting device, when a person existing around the work machine is speaking.

2. The work machine according to claim 1, wherein the control device reports the direction of speech by one or more of methods including displaying the direction of speech on a display device; outputting the sound from a sound output device corresponding to the direction of speech among a plurality of the sound output devices provided inside a driving room; lighting a light source corresponding to the direction of speech among a plurality of the light sources; and vibrating an operation device corresponding to the direction of speech among a plurality of the operation devices.

3. The work machine according to claim 2, wherein the control device displays information indicating the direction of speech on the display device in a display mode that is different from that of information indicating another direction, with reference to display information indicating the work machine.

4. The work machine according to claim 3, wherein the upper turning body is further provided with an imaging device, and the control device displays a bird's-eye view image indicating a surrounding area of the work machine from above the work machine based on image information captured by the imaging device, and displays the information indicating the direction of speech in the display mode that is different from that of the information indicating the other direction, with reference to the display information indicating the work machine included in the bird's-eye view image.

5. The work machine according to claim 2, wherein the upper turning body is further provided with a plurality of imaging devices, and the control device displays image information captured by the plurality of imaging devices on the display device with reference to the work machine, in such a manner that the image information capturing the direction of speech is displayed in a display mode different from that of the image information of another direction.

6. The work machine according to claim 1, wherein the upper turning body is further provided with an imaging device, and the control device corrects the direction of speech based on a direction of a person detected from image information captured by the imaging device.

7. The work machine according to claim 1, wherein the upper turning body is further provided with an imaging device, and the control device prevents reporting of the direction of speech when a direction of a person detected from image information captured by the imaging device is different from the direction of speech.

8. The work machine according to claim 1, wherein a plurality of the sound collecting devices are provided, and each of the plurality of sound collecting devices is provided at a different position of the upper turning body so as to be able to detect the direction of the sound around the work machine.

9. The work machine according to claim 8, wherein the control device prevents a sound generated from the work machine based on a sound signal input from the plurality of sound collecting devices.

10. A control system for a work machine comprising: the work machine including: a lower traveling body; an upper turning body mounted on the lower traveling body so as to freely turn; and a sound collecting device provided on the upper turning body so as to be able to detect a direction of a sound around the work machine, and a control device configured to report a direction of speech based on the direction of the sound detected by the sound collecting device, when a person existing around the work machine is speaking.